Medicine, Health Care and Philosophy (2019) 22:397–405
https://doi.org/10.1007/s11019-018-09880-5

SCIENTIFIC CONTRIBUTION

Enhancement, hybris, and solidarity: a critical analysis of Sandel's *The Case Against Perfection*

Ruud ter Meulen[1,2]

Published online: 23 January 2019
© Springer Nature B.V. 2019

Abstract
This article presents a critical analysis of the views of Michael Sandel on human enhancement in his book *The Case Against Perfection* (2007). Sandel argues that the use of biotechnologies for human enhancement is driven by a will to mastery or *hybris*, leading to an 'explosion of responsibility' and a disappearance of solidarity. I argue that Sandel is using a traditional concept of solidarity which leaves little room for individual differences and which is difficult to reconcile with the modern trend towards individual autonomy and cultural heterogeneity. With reference to the sociology of Giddens, I argue that the 'explosion of responsibility' can be considered an expression of the insecurity in modern society and the need for *reflexivity* in personal and social life. I argue that the need for reflexivity is a driving social and cultural force behind the interest in human enhancement, and that this trend can go together with a commitment to take care of the needs of others. I agree with Sandel that there is a risk that human enhancement may develop into 'hybris', particularly when enhancement is driven by scientism and a naturalist reduction of human values to neurological processes. Enhancement based on scientism will threaten solidarity defined by Honneth as an interactive relationship aimed at the mutual recognition of individual identities in relation to a shared value horizon. I will call for a reflective approach in which human enhancement technologies are explored in the context of a 'system of reference'. Such a reflective approach will help to protect against the hybris of scientism and to maintain solidarity with vulnerable groups.

Keywords Ethics · Human enhancement · Solidarity · Reflexivity · Sandel · Honneth

According to some philosophers, solidarity is disappearing from our society due to the desire of individuals to enhance their physical and mental capacities beyond the normal range (Lev 2011; ter Meulen 2017b). This negative view of human enhancement can particularly be found in the publications of communitarian philosophers, most of all in the work of Michael Sandel. In his book *The Case Against Perfection* (2007) Sandel argues that, because of the idea of human enhancement, we have become *too* responsible for our own fate. Instead of solidarity, which is based on the feeling of a common fate, enhancement technologies will shift responsibility for one's fate from society to the individual, thereby weakening important social values, particularly the responsibility for those with less talents and capabilities (Sandel 2007).

In this article I will argue that Sandel is applying a traditional concept of solidarity in which the interests of the group prevail over the autonomy of the individual. This interpretation of solidarity argues that the personal development and flourishing of an individual is strongly dependent on their participation in the group. This model of solidarity is challenged by modern sociological theories which try to reconcile solidarity with the increasing emphasis on individual autonomy and personal responsibility. The individualization of society is a process which cannot be turned around by falling back on nostalgic ideas of the solidarity of the small community as happens in some accounts of communitarianism, including Sandel's account. In fact, the individualization of society is an important social context for the emergence of enhancement technologies, particularly in the field of cognitive and emotional functioning. I will refer to the work of the sociologist Anthony Giddens to argue that the so-called 'explosion of responsibility' (Sandel) can

✉ Ruud ter Meulen
r.termeulen@bristol.ac.uk

1 Bristol Medical School, University of Bristol, Canynge Hall, 39 Whatley Road, Bristol BS8 2PS, UK

2 Dieren, The Netherlands

be considered as an expression of the insecurity in modern society and the need for *reflexivity* in personal and social life (Giddens 1991).

According to Sandel human enhancement is the result of the drive to master the world, or what he calls a tendency to *hybris* which can be translated as 'overconfidence' or 'pride before the fall'. I argue that there is indeed a risk that human enhancement may develop into 'hybris', particularly when enhancement becomes part of scientism, that is the ideology that all aspects of human life and social life, including the sphere of values, can be explained by the methods of the sciences, particularly the neurosciences. This worldview will result in a naturalist reduction of human values and mental functioning to neurological processes ('we are our brain') and the claim that all aspects of human existence can be 'enhanced' by scientific methods and biotechnologies. I will argue that enhancement based on scientism will threaten solidarity as defined by Axel Honneth as an interactive relationship aimed at the mutual recognition of individual identities in relation to a shared value horizon (Honneth 1995). In this article I call for a reflective approach in which human enhancement technologies are explored and valued in relation to a set of shared values or 'system of reference' which should include important values like solidarity and the protection of the dignity. This reflective approach may help to protect individuals and society against the hybris of scientism and to protect the interests of vulnerable groups and individuals.

The case against perfection

In his book *The Case Against Perfection. Ethics in the Age of Genetic Engineering* (2007) the philosopher Michael Sandel wants to present an ethic of human enhancement that will address fundamental ethical and philosophical questions about the use of genetic technologies (and other biotechnologies) to redesign our nature. Genetic technologies may not only help to treat and prevent a host of debilitating diseases, they may also enable us 'to enhance our muscles, memories, and moods; to choose the sex, height and other genetic traits' of our children; to improve our physical and cognitive capacities; to make ourselves "better than well" (Sandel 2007). According to Sandel, most people find this development 'disquieting', but find it also difficult to articulate the source of their moral unease. It seems that the traditional moral discourse of personal autonomy, fairness and individual rights 'does not equip us to address the most difficult questions of the efforts to enhance our children and our own physical and mental capacities. These 'hardest' and 'unavoidable' questions are questions about 'the status of our human nature and our stance towards the given world' (Sandel 2007). It is not enough to say that the use of genetics for human enhancement are threatening human nature and dignity: we should also argue *how* these practices are doing this and *why* they are causing so much moral distress.

The answer to these questions according to Sandel must be sought in three 'key features of our moral landscape' which he defines as humility, responsibility and solidarity. Humility is the attitude to accept the worlds as it is, meaning that one should be open for the things that happen to us (what Sandel calls 'the unbidden'): 'It invites us to abide the unexpected, to live with dissonance, to reign in the impulse to control.' Humility means acceptance of our own genetic inheritance and reigning in the drive to control the genetic make-up of our children. Humility towards our talents and the feeling of indebtedness to what has been given to us will restrain our tendency toward 'hubris'. The word 'hubris' stems from the ancient Greek word *hybris* which means foolhardiness, overconfidence, or 'pride before the fall'. A well-known example in Greek mythology is the person of Prometheus who was punished for his wish to change the human condition by using technology. Zeus, the Chief of all Greek gods, was so angry by Prometheus' hybris that he nailed him to a rock where the eagle Ethon came each day to pick his liver (Prometheus, who was an immortal, was finally liberated from his predicament by Heracles as one of his eleven works).

According to Sandel, mankind is now acting in the same Promethean spirit: we are trying to be the 'masters of our world', imposing our will on what has been given to us. This Promethean vision of human liberty and power is 'contagious', but also essentially 'flawed': it 'threatens to banish our appreciation of life as a gift, and to leave us with nothing to affirm or behold outside our own will'. This denial of the 'giftedness' of our existence is leading to an 'explosion of responsibility': 'Parents have become responsible for choosing, or failing to choose, the right traits for their children. Athletes become responsible for acquiring, or failing to acquire, the talents that will help their team win.' In the end, this Promethean responsibility will come to a fall: it is not leading to more power, but to a deep form of disempowerment: 'It distracts us from reflecting critically on the world and deadens the impulse to social and political improvement.'

The 'explosion of responsibility' has not only come to the expense of an attitude of acceptance of our limitations, but also to a decline of our motivation toward solidarity. The more we are aware of the gifted and contingent nature of our lot, the more reason there is to share our fate with others. However, when we replace chance with choice, 'the gifted character of human powers will recede, and with it, perhaps, our capacity to see ourselves as sharing a common fate'. The obligation of the successful in our society towards the least advantaged, is essentially based on the notion of giftedness. When the flourishing of the successful is seen

as the result of a natural gift, and not of their own doing, it would be wrong to assume that they would be entitled to all the fruit of their privileged position. However, when society would not be based on contingency, but on personal choice toward enhancement, the gifted nature of our talents would disappear as well as our feeling of sharing a common fate. The successful would see themselves as virtuous and responsible for their own success, while the least advantaged would develop into an 'underclass' of unfit individuals who cannot count on our solidarity or any financial compensation.

The meaning of solidarity

Sandel's argument can be summarized as follows: the (increased) use of genetics and other biotechnologies for human enhancement will strengthen our desire to mastery and will decrease our respect for the gifted nature of our existence. This will lead to a denial of the sense of a common fate and a reinforcement of individual responsibility and the importance of personal merit at the expense of solidarity with the least advantaged in our society. In other words: the denial of humility will lead to an exaggerated responsibility which will result in a decline of solidarity. In this article, these three elements will be further examined, to start with the concept of solidarity.

Solidarity is a multi-layered concept that has origins in various disciplines, particularly in sociology, political science and philosophy (ter Meulen 2017a). The word 'solidarity' was originally used as a political and moral concept in France in the middle of the nineteenth century. In this period social reformists and utopian socialists called for an improvement in the conditions of the working classes by a restructuring of society. The idea of solidarity stems from the legal domain where it meant that members of a family are jointly responsible for the debts of each of its members. Social reformers and theorists used this legal concept to indicate that individuals live in social relationships and that, because of this relationship, they have a moral duty to support each other. The idea of solidarity as mutual support became a major social principle in various social policies, not just in France but in various countries on the European continent. The German Chancellor Bismarck was the first to introduce an insurance-based system of access to health care, social support, compensation for illness and loss of work, and pension at old age. This system of social security was explicitly guided by the idea of solidarity, interpreted as the 'strong' helping the 'weak'. Solidarity in many European countries means that individuals make a financial contribution to a collective system of social security based on public insurances.

At the same time, solidarity became a central concept in the emerging discipline of sociology. In sociology, solidarity refers to the degree of *social cohesion* in a society, particularly the extent to which individuals are prepared to give up their own direct interests in favour of the interests of the group or society in general. The most important sociological theory of solidarity was developed by Emile Durkheim in his book *De la division du travail* (2014 (1893)). Durkheim made a distinction between what he called *mechanical* and *organic* solidarity. 'Mechanical' solidarity describes the social structure of primitive societies with a 'conscience collective'. The commitment to support the collective goals and interests in such a society is self-evident and does not need any justification or reflection by the members of that society ('mechanical'). In modern society, on the contrary, the social cooperation is based on 'organic' solidarity: because of the individualisation of society and the division of labour, social cooperation between the members of modern society is not self-evident and needs to be enforced by the state. Durkheim argued that, because of the individualisation of society, individuals should agree voluntarily with the solidarity enforced by the state.

Solidarity in many European welfare state can be described as an 'organic' type of solidarity: it is supported by individuals because they have an interest to do so ('enlightened self-interest'). It is often called 'interest solidarity': This type of solidarity can be considered as 'weak' in comparison with the 'strong' solidarity of mechanical solidarity in pre-modern societies. The weakness has to do with the tension between obedience to the state and the willingness of the individuals to make a voluntary contribution to the system of solidarity. If individuals are becoming more autonomous while living in an ever more heterogeneous society, there is a concern whether they stay prepared to share benefits and burdens, like paying taxes for the common good and to support the other (Lukes 2014). In a society drifting apart due to the emphasis on individual autonomy and responsibility, cultural diversity and heterogeneity of life-styles, solidarity can become fragile and difficult to organize. In such situations, individuals need to be convinced of the importance of solidarity by appealing to their own interests (Taylor-Gooby 2001).

Solidarity and human enhancement

It is not clear to which extent 'interest solidarity' will be affected by a widespread use of enhancement technologies. As this type of solidarity is based on enlightened self-interest, enhancement technologies might just as well help to strengthen it by enlightening the individuals of their personal interests. An example is cognitive enhancement which tries to improve cognitive functions in the field of memory, attention and learning by pharmacological drugs as well as invasive and non-invasive brain technologies. One

could imagine that the use of these technologies may help individuals to have a better perception of the importance of solidarity, including the personal benefits and interests to cooperate. However, it may also happen that improved insight may lead to diminishing support for solidarity: individuals may get the view that the benefits may not be so great, for example because of the lack of resources or diminishing quality of the health services. In that case they may drop their support as they believe that they do not get a good return on their 'investment' (meaning their financial contribution to the public health insurance system).

The interpretation of solidarity as 'interest solidarity' is a sociological or *descriptive* statement based on empirical observations about human cooperation in modern society. It is a type of solidarity which is meant to cope with the perception of the risks of ill health, disability, illness, unemployment and other perils of modern society. It says nothing about solidarity as a *normative* concept, neither about the moral claims that could be based on such a concept. The normative approach to solidarity has to do with the concern for the well-being of the other in personal relationships (like caring relationships) and the position of the vulnerable and dependent people in society at large. This type of solidarity calls for responsibility for the other by direct personal support or providing specific types of health care and social support for those individuals who are not able to take care of themselves by loss of income or serious mental or physical disabilities. This is sometimes called 'humanitarian solidarity', which is often a powerful motive for solidarity in addition to the interest solidarity mentioned above (ter Meulen 2017a). The concerns of Sandel, as expressed in *The Case Against Perfection* have particularly to do with this 'humanitarian solidarity': The 'explosion of responsibility' may lead to a disappearance of the idea of the giftedness of our talents and of the sense of a common fate which according to Sandel is the basis for solidarity as commitment with the well-being of the vulnerable other.

However, it is not clear to which extent human enhancement may or will affect this type of solidarity. The idea of *moral enhancement* (Douglas 2011; Savulescu and Persson 2008) calls for pharmacological interventions in individuals to improve moral and social capacities like empathy, solidarity, justice and forgiveness as well as willingness to support the other (Lev 2011). Examples of such interventions are pharmacological drugs like oxytocin (which creates a trusting attitude) and antidepressant drugs like Prozac (which creates more friendly behaviour). Some non-pharmacological interventions for cognitive enhancement are also believed to produce such effects (Jotterand 2016). Could enhancement, and particularly *moral enhancement* on a wide scale (as Savulescu and Persson propose), not lead to a strengthening of solidarity? Though some drugs seem to increase an open attitude of individuals, we should not have too high expectations of pharmacological moral enhancement to increase the level of solidarity or moral behaviour in our society. Some drugs can improve moral behaviour, but they can have the opposite effect depending on the context. Oxytocin may help you to trust and co-operate with others in your social group, but it reduces empathy for those outside the group. Basic moral behaviour is to be helpful to others, feel responsible to others, have a sense of solidarity and sense of justice. It is not clear that drugs can ever achieve this. However, they can make us more likeable, more social, less aggressive, more open attitude to other people. It is more likely that they will be more effective in prevention and cure, for example for people with psychopathic tendencies (Jotterand 2016).

Communitarian critique of human enhancement

The comments by Michael Sandel about the negative impact of human enhancement on solidarity are part of general communitarian concerns about the individualisation of society. This sociological term describes the dissolution of traditional bonds and communities which in the past tied people together. It is a process that has been going on for several centuries and has, in modern times, resulted in a disembodiment of the individual as well as instability and changeability of social and personal relations (ter Meulen et al. 2001).

Communitarian authors, including Michael Sandel, complain the loss of social ties and communities which they regard as a morally doubtful process. They argue that the individualization of society has led to an anonymization of social life, a narcissistic occupation with the self (Lasch 1979) and lack of social or political commitment (Bellah et al. 1985). They criticize the modernization of contemporary society which has got the features of what the German sociologist Tönnies called *Gesellschaft* (Tönnies 1887). This term refers to a type of society in which relations between individuals are mainly seen as instrumental to the purposes and interest of everyone. This is the situation in modern societies where individuals have reached much autonomy but are alienated from one another and chasing their personal interests only (think of the 'interest solidarity'). Tönnies placed the *Gesellschaft* in opposition to the *Gemeinschaft* which is the state of relations between individuals characterized by strong communal bonds and social ties. This situation is typical for premodern societies but can also be found in villages and small social communities or religious groups in modern times.

For many communitarian philosophers, including Sandel, solidarity is a normative concept that refers to the altruistic and supportive 'Gemeinschaft' of the pre-modern era. They

emphasize the importance of culture and religious groups which they see as constitutive of people's identity (Sandel 1990). Instead of autonomy and individual independence, a lack of communal life will result in a decrease of possibilities for personal development and a lack of autonomy instead. The use of enhancement technologies seems to contribute to this process, as it creates an obsessive pre-occupation with the self with no commitment for the well-being or problems of others.

Sandel and other communitarian thinkers argue that social ties are an important context for the development of identities. They are generally led by the belief that individuals can only reach fulfilment or a valuable life by commitments and relationships in a community (Kymlicka 2002). Individuals are 'embedded' or 'situated' in social roles and practices: they do not decide to enter such practices, but they are inevitably deeply immersed in them. Instead of taking an outsider position, communitarian authors encourage individuals to immerse themselves in practices and communal relationships to realize their own personal good. Their concern is that human enhancement with its strong emphasis on individual improvement, will go at the expense of participating in social and communal life which they regard as an important condition for personal fulfilment and support of those who are vulnerable and unable to participate in society. Human enhancement, and the various biotechnologies this term refers to, is directing individuals away from life in the community and from contributions to the 'common good', which can be conceived as a substantive idea which defines the community's way of life.

This communitarian concern can be heard in the report *Beyond Therapy* of the U.S. President's Council on Bioethics (2003), for example where it is criticizing the use of medical drugs (like antidepressant drugs) to feel better, even though one is not suffering from (diagnosed) mental illness. Individuals taking such medication want to feel 'better than well' (Elliott 2003). The Council argued that due to the wide availability of mood-brightening drugs individuals become so preoccupied with their state of mind 'that they remove themselves increasingly from active participation in civic life, discarding those attachments without which they cannot achieve the happiness they seek and without which the community cannot survive and flourish' (President's Council on Bioethics 2003).

The communitarian approach is often criticised because it offers only limited and rigid categories of identity. Conventional communitarian notions of solidarity seem to ignore the need for autonomous choice by individuals and for expression and recognition of their individualities. However, there are modern accounts of communitarianism which distance themselves from these 'integrationist' (Benhabib 1992) tendencies of communitarianism. They are concerned with the balance between community and autonomy, between individual rights and social responsibilities. They want to explore the conditions under which the common good and individual rights need to be adjusted to each other instead of the assumption that the community is supreme to the interests of the individual. Etzioni for example, criticizes communitarians like Sandel who stress the importance of communities for the development of individuals, but do not explore the 'opposite danger: that a community may be oppressive, authoritarian, and may unduly penetrate the individual' (Etzioni 1998).

Enhancement and reflexive modernity

Because of the individualization of society, traditional norms and knowledge have become less certain and have lost their status as beacons to guide individuals through institutions and the life cycle (Giddens 1991). Individuals have become the organizers and agents of their own life, but without the guidance by traditional norms and institutions. The modern individual must constantly question and revise their living conditions, a process that has been coined by Giddens as *reflexive modernity*. According to Giddens social and individual life in modern times are characterized by reflexivity, meaning that these practices are constantly examined and reformed in the light of incoming information about these practices, thus constantly changing the character of these practices (Giddens 1990). In modern times, nothing should be taken for granted and everything is in constant revision and examination.

The process of modernization and reflexivity has important consequences for individual lives and for the constitution and formation of identities. Identities are not fixed or based any more in traditional social patterns but are formed and redefined in a constant process of revision and reflection. Modernity is focused on the questions 'who shall I be?' and 'how shall I live?' (Giddens 1991). People are searching for their identity or authenticity, comparing various life styles with each other and trying to make a choice between them. But the choices and the answers to the question of personal identity is the responsibility of the individual him or herself, using his or her reflexive capacities to revise and rewrite their personal identity. Individuals need to learn to become the centres of their own action and of the management of risks in their personal, physical and social environment. Individual biographies have become *self-reflexive* as they have become self-produced in a context where nothing is fixed or determined.

Individuals are enjoying in a lot of freedom, but they also feel an increasing burden of responsibility: personal and social life, biographies and careers, are under constant revision and examination. At the same time individuals are confronted with increasing social and cultural expectations

regarding their performance in work, personal relations and leisure activities (Munz 2015). There is always the uncertainty of having made the right choice. One can indeed speak of an 'explosion of responsibility' (Sandel). However, this is not the result of a drive to mastery and control of our human fate, but of the increasing role of reflexivity and the search for identity and social recognition in modern societies. Human enhancement is a tool for individuals to deal with the uncertainty in their lives and with the responsibility to continuously reflect and improve their lives. Elliott describes this search for authenticity by individuals who are embracing all kinds of enhancement technologies to feel more authentic, more 'being themselves' (Elliott 2003). According to Elliott, 'many people today have the sense that an authentic life is somehow a higher life, a more fulfilled life and that if they do not discover a path that is true to their own aspirations and talents, they are missing out on what life could be' (Elliott 2011).

It can be questioned whether individualisation, and human enhancement, will inevitably result in an impoverishment of individual and communal life. Though many individuals in modern society may struggle with social alienation or indulge in private, consumerist withdrawal from social life, individualisation is not necessarily leading to wrongful and doubtful consequences. Individualization can lead to social and political commitments based on personal reflexivity and autonomous choice instead of the impersonal and obligatory character of such commitments in the past. It can have a positive connotation of personal development, self-realization and emancipation of traditional social ties of religion, small communities, family and class. Sandel criticises human enhancement as it reinforces the idea of personal responsibility instead of solidarity based on a shared fate or sense of luck. However, a greater emphasis on personal responsibility, even an 'explosion of responsibility', can still go together with and even contribute to a reflexive 'ethics of commitment' towards the well-being of others.

Solidarity and recognition of the other

Sandel's concerns, as expressed in *The Case Against Perfection*, are about the role of solidarity as a social value, meaning solidarity which is aimed at the protection and inclusion in our society of individuals who are in a vulnerable or a less advantaged position. However, there is also another type of solidarity, that is the solidarity which plays an important role at the level of personal relationships. In this type of solidarity, the individual realizes him or her own self in a relation of mutual recognition. According to the philosopher Honneth (1995) this type of solidarity is typical for Hegel's 'ethical life' (*Sittlichkeit*) which is based on a relational and contextual approach of individual development. Hegel's 'ethical life' must be distinguished from Kant's concept of 'morality' (*Moralität*) in which the dignity of persons and the promotion of self-respect plays a central role. While the concept of dignity respects and protects human rights, solidarity promotes the formation of identity and self-esteem under the condition of mutual relatedness and fundamental interdependency of individuals (Honneth 1995). Both ethical traditions are important and complimentary in regard with the recognition of individuals and the development identities (Honneth 1995).

Where law provides a medium for the recognition of universal rights, solidarity requires a different medium to express the characteristic differences between individuals. Honneth argues that solidarity requires the articulation at a societal level of a 'framework for orientation', in which 'those ethical values and goals are articulated that, taken together, comprise the cultural self-understanding of a society' (Honneth 1995). Such a framework of 'horizon of values' can serve as a system of reference for the appraisal of individual personality features, particularly the degree to which they each in their own way contribute to the realization of societal goods. Solidarity in Honneth's work then means an interactive relationship in which individuals mutually sympathize with each other's different ways of life because they esteem each other in relation to a shared value horizon. While the group is the first instance for such recognition of individual differences, and for the self-esteem resulting from it, such solidarity can be extended to other members of society (Honneth 1995).

In a comparable way, Dean argues that, solidarity should be understood as based on a communicative practice in which individuals create a 'we' in a *reflective* process (Dean 1996). Solidarity can never be viewed as fixed or as given: achieving solidarity means that we open the membership of a group or community to dialogue and communicative reflection on social values and expectations. Dean argues that by reflecting on shared values, solidarity can become an *inclusive* solidarity which promotes self-esteem by way of solidarity and self-respect. Dean calls this *reflective solidarity*, meaning a solidarity which strengthens social ties by a reflection on social expectations and recognition of personal identity. This is a different interpretation of the relation between individuality and the role of social communities as proposed in traditional (including Sandel's) accounts of communitarianism.

Scientism, hybris and the community of values

An important question is how human enhancement may affect this type of solidarity in which mutual recognition and the promotion of self-esteem are central. One could argue

that certain types of human enhancement, particularly mood enhancement, cognitive enhancement and moral enhancement, could make a positive contribution to an attitude of openness and respect of the other. However, human enhancement can turn into a threat for this type of solidarity, when it becomes driven by scientism and naturalism. Scientism considers the methods of the natural sciences as the only source of valuable knowledge of the world and that solely these methods can provide reliable knowledge not only about nature, but also about man and society. Scientism leads to a reductionist approach in the knowledge of the social world, excluding other kinds of knowledge like historical, philosophical and cultural viewpoints. Scientism is accompanied by a naturalistic worldview, according to which reality is exhausted by nature and that there is nothing supernatural or metaphysical in our world (Reiner 2011). According to naturalism, the methods of the natural sciences are the one and only way to investigate nature and human existence.

Many efforts towards enhancement of human capacities are driven by such reductionist views. A good example is, what Reiner calls 'neuroessentialism', which is the position that 'for all intents and purposes, we are our brains' (Reiner 2011). Neuroessentialism is a reductionist, naturalist approach in so far as it holds the position that mental states, behaviour, notions of self, and personal identity can be reduced to neurobiology (Jotterand 2016). In the same way as genetics has been used to support the idea that 'we are our genes', neuroessentialists claim that 'we are our brains'. Interventions in the brain, by pharmacological means or invasive procedures, are the best way not only to treat disorders, but also to enhance the mental capacities and social and moral life of human individuals. This is for example the case in the efforts towards moral enhancement, which regards the application of drugs and other methods as the best way to improve relationships between individuals. This reductionist approach is problematic, because it limits morality to moral emotions ('we are our brains'); it does not consider that moral agents develop their understandings of notions of the good in interactions with significant others and in the exploration of the values which they share (or not share). This *community of values* (ter Meulen 2010) can never be reduced to processes in the brain or 'enhanced' via the use of drugs or other biotechnologies.

As Honneth argues, solidarity means an interactive relationship in which individuals mutually sympathize with each other's different ways of life because they esteem each other in relation to a shared value horizon. This type of solidarity would become impossible when moral life and the community of values are reduced to a neurological process which can be simply enhanced. This is not only a simplistic view of human existence, it is also a serious threat to the moral life and solidarity in contemporary society.

According to Sandel, our efforts towards enhancement are a sign of hybris as it reflects our desire to be the 'masters of our world', imposing our will on what has been given to us ('the given'). Though this mastery may indeed drive some of the enhancement movement, a more fundamental threat is represented by the naturalist drive to reduce values to identifiable brain processes. This scientism will threaten solidarity as mutual recognition of identities as it will annihilate the importance of the community of values on which this recognition is based. Naturalism, or more specifically the *naturalistic fallacy* (Hume), has been held responsible for an instrumental view of human existence and of efforts to improve the 'quality' of humanity and eugenics (Brüne 2007). It is called a 'fallacy' as it tries to reduce moral life to biological processes, which could be 'enhanced' by pharmacological drugs.

Jotterand makes a helpful distinction between moral *capacity* and moral *content*. Moral capacity refers to one's ability or disposition to respond morally and 'involves the motivational, cognitive, and affective mental processes determining how one behave when confronted with moral dilemmas' (Jotterand 2016). Moral content constitutes 'the set of particular beliefs, values and ideas shaped by the environment, cultural, and historical factors in addition to rational and moral deliberation and moral theorizing' (idem). A reductionist approach limits moral content to moral emotions and capacities and does not consider that moral agents develop shared understandings of the right and the good that shape their behaviour and identities. This is the social context where personal identities and moral agency are formed, but which cannot be reduced to these capacities. This is the community of values or the 'value horizon' which is the point of reference for the development of solidarity in personal relationships. One can speak of *societal* solidarity to the extent that every member of society is able to esteem himself or herself in relation with this shared horizon of values (Honneth 1995). Any effort to reduce this sphere of values will diminish the development of solidarity and the potential to develop identities and self-respect.

Enhancement, responsibility and solidarity

In *The Case Against Perfection* Sandel argues that the use of biotechnologies for human enhancement is driven by a will to mastery leading to an 'explosion of responsibility' and a disappearance of solidarity. I have argued that Sandel is using a traditional communitarian view of solidarity in which individual personal development is strongly dependent on participation in the life of the community but leaves little room for individual differences. This approach is difficult to reconcile with the modern trend towards individual

autonomy and cultural heterogeneity, including the efforts to enhance human capacities by technological means.

According to Giddens social and individual life in modern times are characterized by *reflexivity*, meaning that these practices are constantly examined and reformed in the light of incoming information about these practices, thus constantly changing the character of these practices (Giddens 1990). In modern times, identities are not fixed or based any more in traditional social patterns but are formed and redefined in a constant process of revision and reflection. Individuals have become the centres of their own action while their biographies have become *self-reflexive*, produced in a context where nothing is certain. The efforts towards human enhancement must be situated in this modern context: human enhancement technologies are a tool for individuals to deal with the uncertainty of modern life and with the responsibility to reflect and change their lives towards perfection.

Sandel criticises human enhancement as it reinforces the idea of personal responsibility instead of solidarity based on a shared fate or sense of luck. I have argued that a greater emphasis on personal responsibility, even an 'explosion of responsibility', can still go together with and even contribute to a reflexive 'ethics of commitment' towards the well-being of others. Such an approach is in line with modern theories of communitarianism which try to find a balance between autonomy and solidarity. Modern theories of solidarity call for a reflective approach in which solidarity is directed towards mutual recognition against the background of a 'system of reference' or 'horizon of values' (Taylor 1991). By reflecting on shared values, solidarity can become an *inclusive* solidarity which promotes self-esteem by way of solidarity and self-respect.

Unlike Sandel, I believe that human enhancement has the potential to lead to a greater social commitment of individuals and to an inclusive solidarity in our individualized society. Cognitive and moral enhancement technologies could help the capacity of individuals towards a better understanding and openness in respect with each other's perspectives and interests. However, I share Sandel's concerns that human enhancement may risk developing into a *hybris*. I have argued that this might be the case when enhancement is driven by scientism and naturalism: these metaphysical attitudes argue that social values and the human mind must be analyzed as natural phenomena and that they should be considered as the mere product of neurological processes ('we are our brain'). These reductionist approaches deny the independent existence of a shared 'horizon of values' which is conditional for solidarity between individuals and for the development of their identities and self-esteem. Solidarity as mutual recognition would be severely hindered or even annihilated when moral life and the community of values are reduced to a brain process which can be simply enhanced. This would indeed be a serious threat to the moral life and solidarity in contemporary society.

Conclusion

The introduction and use of enhancement technologies should be guided by a shared understanding against the background of a shared horizon of values, which should include an understanding of human dignity, solidarity and responsibility. When enhancement is driven by scientism, human beings are at risk to be entirely controlled by technological devices stripping them of any individuality and the capacity to develop meaningful relationships with others. However, such a de-humanization is not a necessary consequence of enhancement technologies. Human enhancement should be integrated in a reflective process, in which the meaning and impact of these technologies on individual and social life should be assessed against the background of a 'horizon of values'. This is, what I would call, *reflective* enhancement: it is a type of enhancement that might help human beings to connect with other human beings and to integrate technology in their own narrative understanding of themselves. Reflective enhancement could help individuals to develop human relationships which are essential for the recognition of their individuality as well as to foster a commitment to support those who are at risk to be left behind. Enhancement technologies do not inevitable lead to a diminishing of solidarity as an important principle for human recognition and social support but may support such values if introduced in a *reflective* way.

Acknowledgement The author is grateful to the Brocher Foundation for granting him a one-month residential fellowship at their premises in Hermance, Geneva, in October 2018, during which this article was written.

References

Bellah, Robert, Richard Madsen, William Sullivan, Ann Swidler, and Steven Tipton. 1985. *Habits of the Heart. Individualism and Commitment in American life*. New York: Harper and Row.

Benhabib, Seyla. 1992. *Situating the Self. Gender, Community and Postmodernism in Contemporary Ethics*. New York: Routledge.

Brüne, Martin. 2007. On human self-domestication, psychiatry, and eugenics. *Philosophy, Ethics, and Humanities in Medicine*. 2: 21. https://doi.org/10.1186/1747-5341-2-21.

Dean, Jody. 1996. *Solidarity of strangers: Feminism after identity politics*. Berkeley: University of California Press.

Douglas, Tom. 2011. Moral enhancement. In *Enhancing Human Capacities*, ed. Julian Savulescu, Ruud ter Meulen, and Guy Kahane, 467–485. Chicester: Wiley-Blackwell.

Durkheim, Emile. 2014 (1893). The division of labor in society. Edited and with a new introduction by Steve Lukes. Translation by W.D. and Hall. New York: Free Press.

Elliott, Carl. 2003. *Better than Well. American Medicine Meets the American Dream*. New York: W.W. Norton & Company.

Elliott, Carl. 2011. Enhancement technologies and the modern self. *The Journal of Medicine and Philosophy* 36: 364–374.

Etzioni, Amitai. 1998. Introduction. In *The Essential Communitarian Reader. The Essential Communitarian Reader*, ed. Amitai Etzioni, ix–xxiv. Lanham: Rowham & Littlefield Publishers.

Giddens, Anthony. 1990. *The Consequences of Modernity*. Cambridge: Polity Press.

Giddens, Anthony. 1991. *Modernity and Self-Identity. Self and Society in the Late Modern Age*. Cambridge, MA: Polity Press.

Honneth, Axel. 1995. *The Struggle for Recognition. The Moral Grammar of Social Conflicts*. Cambridge, MA: Polity Press.

Jotterand, Fabrice. 2016. Moral enhancement, neuroessentialism, and moral content. In *Cognitive Enhancement: Ethical and Policy Implications in International Perspectives*. Ed. Fabrice Jotterand and Veljko Dubljevic. Oxford Scholarship Online. https://doi.org/10.1093/acprof:oso/9780199396818.0001.

Kymlicka, Will. 2002. *Contemporary Political Philosophy. An Introduction*. 2nd ed. Oxford: Oxford University Press.

Lasch, Christopher. 1979. *The Culture of Narcissism. American Life in An Age of Diminishing Expectations*. New York: Warner Books.

Lev, Ori. 2011. Will biomedical enhancements undermine solidarity, responsibility, equaliy and autonomy? *Bioethics* 25: 177–184.

Lukes, Steven. 2014. Introduction to this edition. In *Emile Durkheim, The Division of Labor in Society*, xxv–xlvi. New York: Free Press.

Meulen, Ruud ter. 2010. Dignity, posthumanism and the community of values. Answer to Fabrice Jotterand and Nick Bostrom. *American Journal of Bioethics* 10: 69–70.

Meulen, Ruud ter. 2017a. *Solidarity and Justice in Health and Social Care*. Cambridge: Cambridge University Press.

Meulen, Ruud ter. 2017b. Is human enhancement a threat to solidarity? *Acta Philosophica* 26: 307–322.

Meulen, Ruud ter, Wil Arts, and Ruud Muffels. 2001. Solidarity, health and social care in Europe. Introduction to the volume. In *Solidarity in Health and Social Care in Europe. Philosophy & Medicine*, ed. Ruud ter Meulen, Wil Arts, Ruud Muffels. Vol. 69, 1–11. Dordrecht: Kluwer Academic Publishers.

Munz, Victor. 2015. Why do we wish to be enhanced? In *Inquiring into Human Enhancement. Interdisciplinary and International Perspectives. Health, Technology and Society*. ed. Simone Bateman, Jean Gayon, Sylvie Allouche, Jerôme Goffette, and Michaele Marzano. Basingstoke: Palgrave MacMillan.

President's Council on Bioethics. 2003. *Beyond Therapy. Biotechnology and the Pursuit of Happiness*. Washington, DC: U.S. Government Printing Office.

Reiner, Peter. 2011. The rise of neuroessentialism. In *Oxford Handbook of Neuroethics*. ed. Judy Illes and Barbara Sahakian. Oxford: Oxford University Press. https://doi.org/10.1093/oxfordhb/9780199570706.013.0049.

Sandel, Michael. 1990. Freedom of conscience or freedom of choice? In *Articles of Faith, Articles of Peace*. ed. J. Hunter, and O. Guiness, 74–92. Washington: Brookings Institute.

Sandel, Michael. 2007. *The Case Against Perfection. Ethics in the Age of Genetic Engineering*. Cambridge, MA: Belknap Press.

Savulescu, Julian and Ingmar Persson. 2008. The perils of cognitive enhancement and the urgent imperative to enhance the moral character of humanity. *Journal of Applied Philosophy* 25: 162–167.

Taylor, Ch. 1991. *The Ethics of Authenticity*. Cambridge, MA: Harvard University Press.

Taylor-Gooby, Peter. 2011. Does risk society erode welfare state solidarity? *Policy & Politics* 39: 147–161.

Tönnies, Ferdinand. 1887. *Gemeinschaft und Gesellschaft*. Leipzig: Fues's Verlag.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Reproduced with permission of copyright owner. Further reproduction prohibited without permission.